

Taylor's Instability 2.3. Vórtices de Taylor

Taylor's instability is observed in a viscous incompressible fluid column contained between two vertical cylinders rotating at different angular velocities, the temperature being assumed to remain uniform.

There occurs a competition between centrifugal forces and viscosity: if we assume that the inner cylinder moves at a higher velocity than the outer one, the fluid close to the inside wall will move outward and replace there the slow moving fluid. At

small angular velocity, one has an ordinary Couette flow where angular momentum is transported from the inner to the outer cylinder by viscosity in the form of stationary annular convective cells. When the angular velocity is gradually increased, this state becomes unstable and toroidal vortices, also called Taylor's vortices, are formed. At higher angular velocities, wavy deformations appear and finally a transition to turbulent motion occurs.

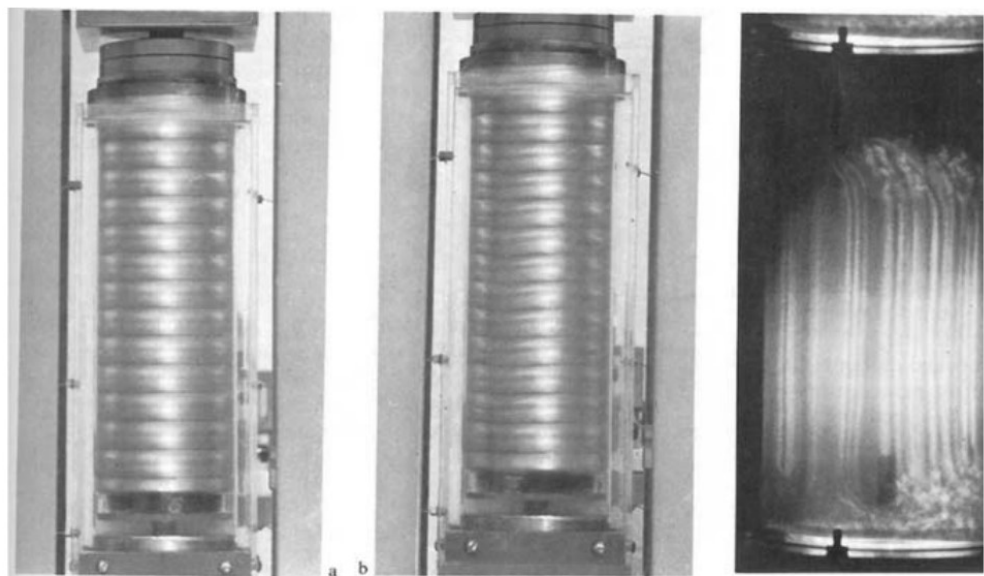
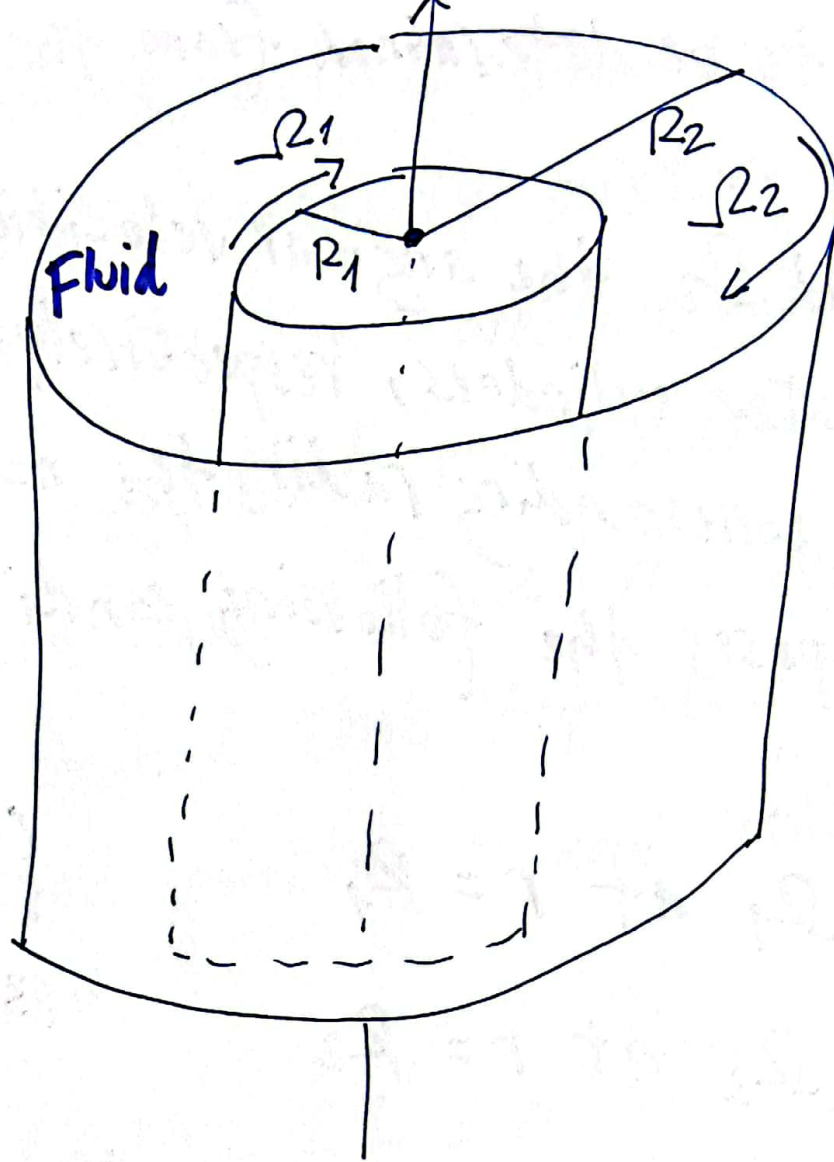


Fig. 6.10 (a) Taylor's vortices, (b) wavy flow, and (c) turbulence (from Coles 1965)



At low angular velocity, the motion is laminar with a velocity field given by

$$v_r = 0, \quad v_\theta = Ar + \frac{B}{r}, \quad v_z = 0 \quad (1)$$

when expressed in cylindrical coordinates r, θ, z , where r is the radial distance to the axis of the cylinders, θ the azimuthal angle, and z the coordinate parallel to the vertical axis, A and B

are constants to be determined from the boundary conditions.

With Ω_1 and Ω_2 the angular velocities of the inner and outer cylinders, respectively, and R_1 and R_2 the corresponding radii, the non-slip condition imposes the following conditions at the walls

$$v_\theta = R_1 \Omega_1 \text{ at } r = R_1 \quad (2)$$

$$v_\theta = R_2 \Omega_2 \text{ at } r = R_2 \quad (3)$$

this allows us to determine the values of the constants A and B , namely

$$A = -\Omega_1 \left(\frac{\Omega^2 - \mu}{1 - \Omega^2} \right) \quad (4)$$

$$B = \Omega_1 \frac{R_1^2 (1 - \mu)}{1 - \Omega^2} \quad (5)$$

$$\text{where } \Omega \equiv \Omega_2 / \Omega_1 \quad (6a)$$

$$\text{and } \mu \equiv R_1 / R_2 \quad (6b)$$

We proceed with a linear stability analysis. Denote the velocity disturbance by u' , v' , w' , and the pressure disturbance by p' ; assuming that the disturbances are small and axisymmetric, i.e. θ independent, it is easily checked that they obey the following linearized continuity and Navier-Stokes equations

$$\frac{1}{r} \frac{\partial}{\partial r} (r u') + \frac{\partial w'}{\partial z} = 0 \quad (7a)$$

$$\frac{\partial u'}{\partial t} - 2v' \frac{v'}{r} = -\frac{\partial}{\partial r} \left(\frac{p'}{\rho} \right) - \nu \left(\nabla^2 u' - \frac{u'}{r^2} \right) \quad (7b)$$

$$\frac{\partial v'}{\partial t} + \left(\frac{\partial v'}{\partial r} + \frac{v'}{r} \right) u' = \nu \left(\nabla^2 v' - \frac{v'}{r^2} \right) \quad (7c)$$

$$\frac{\partial w'}{\partial t} = -\frac{\partial}{\partial z} \left(\frac{p'}{\rho} \right) + \nu \nabla^2 w' \quad (7d)$$

$$\text{where } \nabla^2 \equiv \frac{1}{r} \left(\frac{\partial}{\partial r} \right) + \frac{\partial^2}{\partial r^2} + \frac{\partial^2}{\partial z^2} \quad (8)$$

The pressure disturbance p' is eliminated from relations (7b) and (7d).

Moreover, by supposing that the fluid column is of infinite length, we need only the boundary conditions at R_1 and R_2 , namely

$$u' = v' = w' = 0 \quad (9)$$

According to the normal mode technique, we seek solutions of the form

$$(u', v', w') = [U(r), V(r), W(r)] \exp(ikz + \omega t) \quad (10)$$

To simplify the calculations, we shall make the narrow-gap approximation, which implies that the distance $d = R_2 - R_1$ between the cylinders is much smaller than R_1 . This permits us to put $\mu = 1$ and to omit $1/r$ compared to $\partial/\partial r$, moreover, within the small-gap limit, $A + B/r^2 \approx \Omega_1$ and $A \approx -\Omega_1$.

Substituting (10) in (7), making all the quantities dimensionless, and proceeding by analogy with the Rayleigh-Bénard's problem, we obtain the following relations for the radial and azimuthal velocity amplitudes at the marginal state $\lambda = 0$

$$(D^2 - k^2)^2 U = [1 + (\Omega - 1)Z] V \quad (11)$$

$$(D^2 - k^2) V = -Ta k^2 U \quad (12)$$

where we have introduced the notation $D = d/dZ$ with

$$Z = (r - R_1) / d \quad (13)$$

and

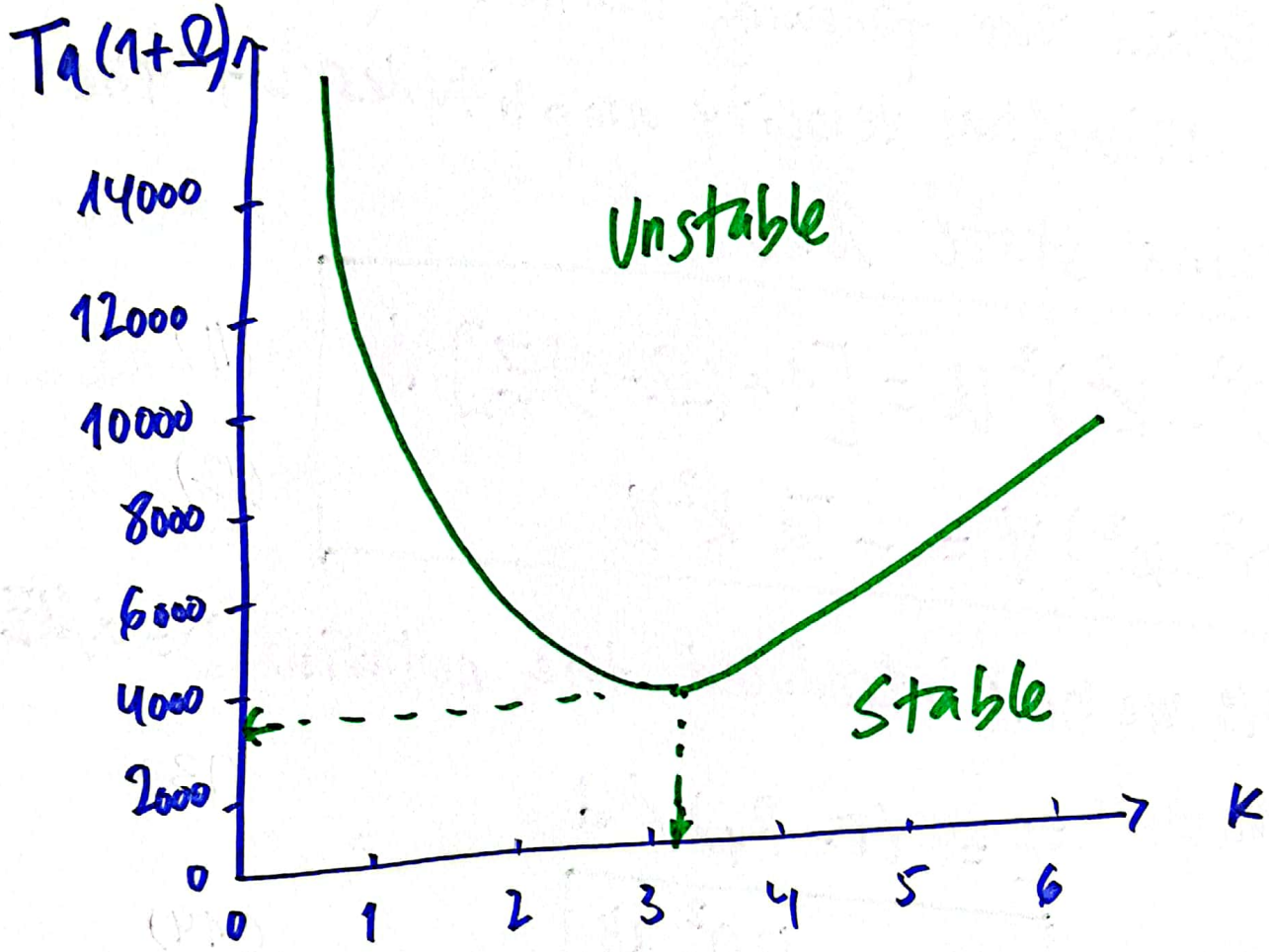
$$Ta = \frac{4\Omega_1^2 d^4}{\nu^2} \quad (14)$$

the relevant boundary conditions are

$$U = DU = V = 0 \quad \text{at } Z = 0 \quad \text{and } Z = 1 \quad (15)$$

The quantity Ta is the dimensionless Taylor number, which is the ratio between the centrifugal forces and the viscous dissipation.

For each particular value of $\Omega = \Omega_2/\Omega_1$, the eigenvalue problem (11) and (12) results in a marginal stability curve (see Figure).



The minimum of the curve gives the critical value $(Ta)_c$ determining the minimal angular velocity at which toroidal axisymmetric vortices will set in.

For $Ta < (Ta)_c$, all the disturbances are dampened and the flow is independent of the z coordinate.

At $Ta = (Ta)_c$, there is a supercritical transition from Couette flow to Taylor's vortices characterized by a critical wave number k_c or a critical wavelength $\lambda_c = \frac{2\pi}{k_c}$ which is detectable on the photographs.

In the small-gap limit, the critical values are given by

$$(Ta)_c = \frac{3430}{(1+\Omega)}$$

$$k_c = 3.12 \quad \text{for } 0 < \Omega < 1$$

The critical Taylor number is a function of Ω and it is interesting to note that, for $\Omega = 1$, one has $(Ta)_c = 1715$, which is very close to the value 1708 obtained for the critical Rayleigh number in the rigid-rigid convection case.

The same remark applies to the critical wave-number whose value $k_c = 3.12$ is comparable to the value $k_c = 3.117$ of the Rayleigh-Bénard problem.

It is therefore clear that the linear stability analysis / problems for Rayleigh-Bénard's and Taylor's instabilities in the narrow-gap approximation are formally identical.